

ME 300:
Engineering Thermodynamics

Example:
Vapor
Compression Cycle



Vapor compression

A refrigerator uses refrigerant R-134a as the working fluid and operates on an ideal vapor compression cycle between 0.14 and 0.8 MPa, with State 1 as a saturated vapor and State 3 as a saturated liquid. If the mass flow rate of the refrigerant is 0.05 kg/s, determine:

- The rate of heat removal from the refrigerated space $\rightarrow {}_4\dot{Q}_1 = \dot{m}(h_1 - h_4) \star$
- The power input to the compressor $\rightarrow {}_1\dot{W}_2 = -\dot{m}(h_2 - h_1)$
- The rate of heat rejection to the environment $\rightarrow {}_2\dot{Q}_3 = \dot{m}(h_3 - h_2)$
- Coefficient of performance

Given : $P_1 = 0.14 \text{ MPa} = P_4$
 $P_2 = 0.8 \text{ MPa} = P_3$
 $x_1 = 1$
 $x_3 = 0$
 $\dot{m} = 0.05 \text{ kg/s}$

Assume : reversible

: S.C.S.

: steady-flow

: quasi-steady

$$\beta = \frac{{}_4\dot{Q}_1}{{}_1\dot{W}_2}$$

Vapor compression

Table A.2 Saturation Properties of Refrigerant R-134a: Pressure Increments

Press., MPa	Temp., T_{sat} , °C	Specific Volume, m ³ /kg		Internal Energy, kJ/kg		Enthalpy, kJ/kg			Entropy, kJ/kg·K	
		Sat. Liquid, v_f	Sat. Vapor, v_g	Sat. Liquid, u_f	Sat. Vapor, u_g	Sat. Liquid, h_f	Evap., h_{fg}	Sat. Vapor, h_g	Sat. Liquid, s_f	Sat. Vapor, s_g
0.06	-36.935	0.00070982	0.31123	151.96	357.27	152.00	223.94	375.94	0.81202	1.7601
0.08	-31.115	0.00071854	0.23755	159.31	360.61	159.37	220.25	379.62	0.84278	1.7528
0.1	-26.361	0.00072593	0.19256	165.37	363.34	165.44	217.16	382.60	0.86756	1.7475
0.12	-22.310	0.00073244	0.16214	170.56	365.67	170.65	214.47	385.12	0.88844	1.7435
0.14	-18.760	0.00073830	0.14015	175.14	367.70	175.24	212.08	387.32	0.90656	1.7402
0.16	-15.588	0.00074368	0.12349	179.25	369.51	179.37	209.90	389.27	0.92262	1.7376
0.18	-12.712	0.00074868	0.11042	183.00	371.15	183.13	207.89	391.02	0.93709	1.7353
0.2	-10.076	0.00075337	0.099877	186.45	372.64	186.60	206.02	392.62	0.95027	1.7334
0.24	-5.3653	0.00076202	0.083906	192.65	375.30	192.83	202.61	395.44	0.97364	1.7303
0.28	-1.2277	0.00076993	0.072360	198.14	377.62	198.35	199.54	397.89	0.99399	1.7278
0.32	2.4768	0.00077727	0.063611	203.09	379.69	203.34	196.70	400.04	1.0121	1.7257
0.36	5.8412	0.00078418	0.056744	207.61	381.54	207.90	194.07	401.97	1.0284	1.7240
0.4	8.9306	0.00079073	0.051207	211.79	383.24	212.11	191.61	403.72	1.0433	1.7226
0.5	15.735	0.00080595	0.041123	221.10	386.91	221.50	185.97	407.47	1.0759	1.7197
0.6	21.572	0.00081998	0.034300	229.19	389.99	229.68	180.89	410.57	1.1037	1.7175
0.7	26.713	0.00083320	0.029365	236.41	392.64	236.99	176.21	413.20	1.1280	1.7156
0.8	31.327	0.00084585	0.025625	242.97	394.96	243.65	171.81	415.46	1.1497	1.7140
0.9	35.526	0.00085811	0.022687	249.01	397.01	249.78	167.65	417.43	1.1695	1.7126
1	39.388	0.00087007	0.020316	254.63	398.85	255.50	163.66	419.16	1.1876	1.7113
1.2	46.315	0.00089351	0.016718	264.88	401.98	265.95	156.09	422.04	1.2201	1.7087
1.4	52.422	0.00091671	0.014110	274.12	404.54	275.40	148.90	424.30	1.2489	1.7062
1.6	57.906	0.00094010	0.012126	282.61	406.64	284.11	141.93	426.04	1.2748	1.7036
1.8	62.895	0.00096400	0.010562	290.52	408.35	292.26	135.10	427.36	1.2987	1.7007
2	67.481	0.00098877	0.0092915	297.98	409.70	299.95	128.33	428.28	1.3209	1.6976
2.5	77.577	0.0010569	0.0069403	315.20	411.65	317.84	111.17	429.01	1.3711	1.6881
3	86.203	0.0011413	0.0052813	331.28	411.49	334.70	92.640	427.34	1.4171	1.6748

States

$$\begin{array}{l} \text{State 1: } P_1 = 0.14 \text{ MPa} \\ \quad \quad \quad X_1 = 1 \end{array} \left. \vphantom{\begin{array}{l} \text{State 1: } P_1 = 0.14 \text{ MPa} \\ \quad \quad \quad X_1 = 1 \end{array}} \right\} \text{A.2} \rightarrow \begin{array}{l} h_1 = h_g = 387.32 \frac{\text{kJ}}{\text{kg}} \\ s_1 = s_g = \underline{1.7402} \frac{\text{kJ}}{\text{kg-K}} \end{array}$$

$$\begin{array}{l} \text{State 2: } P_2 = 0.8 \text{ MPa} \\ \quad \quad \quad s_2 = s_1 = 1.7402 \frac{\text{kJ}}{\text{kg-K}} \end{array} \left. \vphantom{\begin{array}{l} \text{State 2: } P_2 = 0.8 \text{ MPa} \\ \quad \quad \quad s_2 = s_1 = 1.7402 \frac{\text{kJ}}{\text{kg-K}} \end{array}} \right\} \text{A.2} \rightarrow \begin{array}{l} s_2 > s_g \rightarrow \text{vapor} \\ \rightarrow h_2 = 424.59 \frac{\text{kJ}}{\text{kg}} \end{array}$$

$$\begin{array}{l} \text{State 3: } P_3 = 0.8 \text{ MPa} \\ \quad \quad \quad X_3 = 0 \end{array} \left. \vphantom{\begin{array}{l} \text{State 3: } P_3 = 0.8 \text{ MPa} \\ \quad \quad \quad X_3 = 0 \end{array}} \right\} \text{A.2} \rightarrow h_3 = h_f = 243.65 \frac{\text{kJ}}{\text{kg}}$$

$$\begin{array}{l} \text{State 4: } h_4 = h_3 = 243.65 \frac{\text{kJ}}{\text{kg}} \\ \quad \quad \quad P_4 = 0.14 \text{ MPa} \end{array}$$

Work and heat

$${}_1\dot{W}_2 = -\dot{m}(h_2 - h_1) = -0.05(424.59 - 387.32) = -1.86 \text{ kW}$$

$${}_2\dot{Q}_3 = \dot{m}(h_3 - h_2) = 0.05(243.65 - 424.59) = -9.047 \text{ kW}$$

heat rej.
to atm.

$${}_4\dot{Q}_1 = \dot{m}(h_1 - h_4) = 0.05(387.32 - 243.65) = 7.1835 \text{ kW}$$

heat absorbed
from cold space

$$\beta = \frac{{}_4\dot{Q}_1}{{}_1\dot{W}_2} = \frac{7.1835}{1.86} \rightarrow \boxed{\beta = 3.86}$$